CHAPTER 2

MOTION

Introduction

- The study of physics involves the study of matter and its motion
- If the sun, moon, wind and oceans stopped moving, it means that the life and the whole universe would not exist.
- Motion is indeed a *nai'mat* af Allah Ta'ala
- Allah ta'ala makes movement possible.
- He has set celestial bodies in motion. He states

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And Allah Ta'ala subjected the sun, moon– both are in motion for a set period

Types of motion

1. Linear motion- The motion of a body in a straight line.

- A body is said to be in a linear motion if the position/location of the body changes w.r.t frame of reference

2. Circular motion- movement of an object in a circle around a centre point. (along a circumference of a circle.

- Both revolution and rotation are circular motion

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Other types of motion include

- Vibratory or oscillatory motion–this is a to-and-fro movement of an object about a fixed point. E.g pendulum
- Periodic motion– this is any motion that repeats itself at regular intervals of time. Eg. Pendulum of a clock, a pause and heart-beat

***Note**–oscillatory motion will almost certainly be periodic but not all periodic motion will be oscillatory*

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- Random motion- it is a movement in which a an object suddenly changes its nature of motion e.g balloon filled with air executes random motion, insects, Brownian motion of gases

Measuring speed

- Speed is an expression of how fast an object moves.
- An object's speed is the rate of the total distance it travels in a certain amount of time.

Note

- Speed is comparing one quantity to another of the same kind.
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- Units of speed include miles per hour (mph), centimeters per second (cm/sec or cm/s), meters per second (m/sec or m/s), or kilometers per hour (kph or km/hr).
- Measuring speed involves observing the distance an object travels and the amount of time elapsed and then calculating the speed from those observations by dividing distance by time.

Distance, time and speed

- To determine the speed of a moving body we have to make two measurements

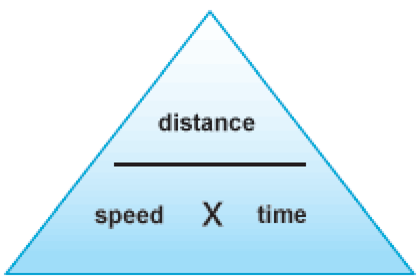
1. The distance travelled by a body between two points
2. The time taken to travel between these two points

Note. The SI unit for speed is **m/s** or **km/hr**

Speed= distance/time

Examples

1. A cyclist in *khaimat al-riyadat* completes a 1500m stage of race in 37.5sec. What was his speed?
2. A spacecraft is orbiting the earth at a steady speed of 8km/s. how long will it take to complete a single orbit, a distance of 40000km?



You may find this triangle useful when rearranging the equation to get:
distance = speed x time
time = distance / speed

Some modes of transport used by al-Dai al-Ajal Syedna Mohammed Burhanuddin

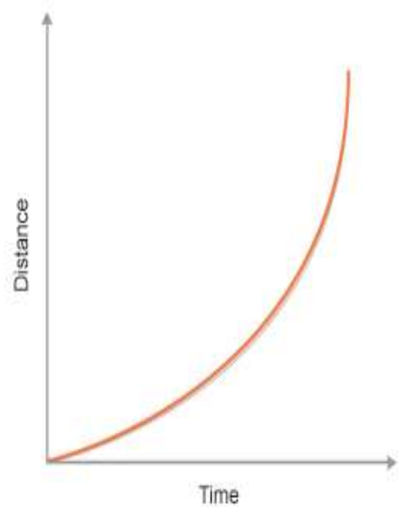
- Concorde
- Green Buick of His Holiness
- Cunard Liner (Queen Elizabeth-II
- Eurostar
- White stretch Mercedes Benz

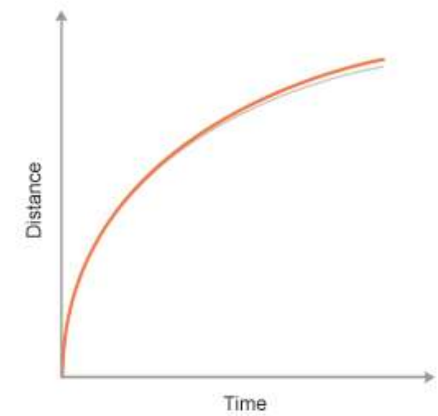
Acceleration

- Acceleration is the rate at which an object changes its speed or velocity
- Suppose you complete the first *shaut* in 5minutes, and second 4minutes, you will say that your speed has increased.
- Also suppose you complete the first *shaut* in 4minutes, and second in 5minutes, you will say that your speed has decreased.
- We say that an object **accelerates** when its speed increases and **decelerates** when its speed decreases

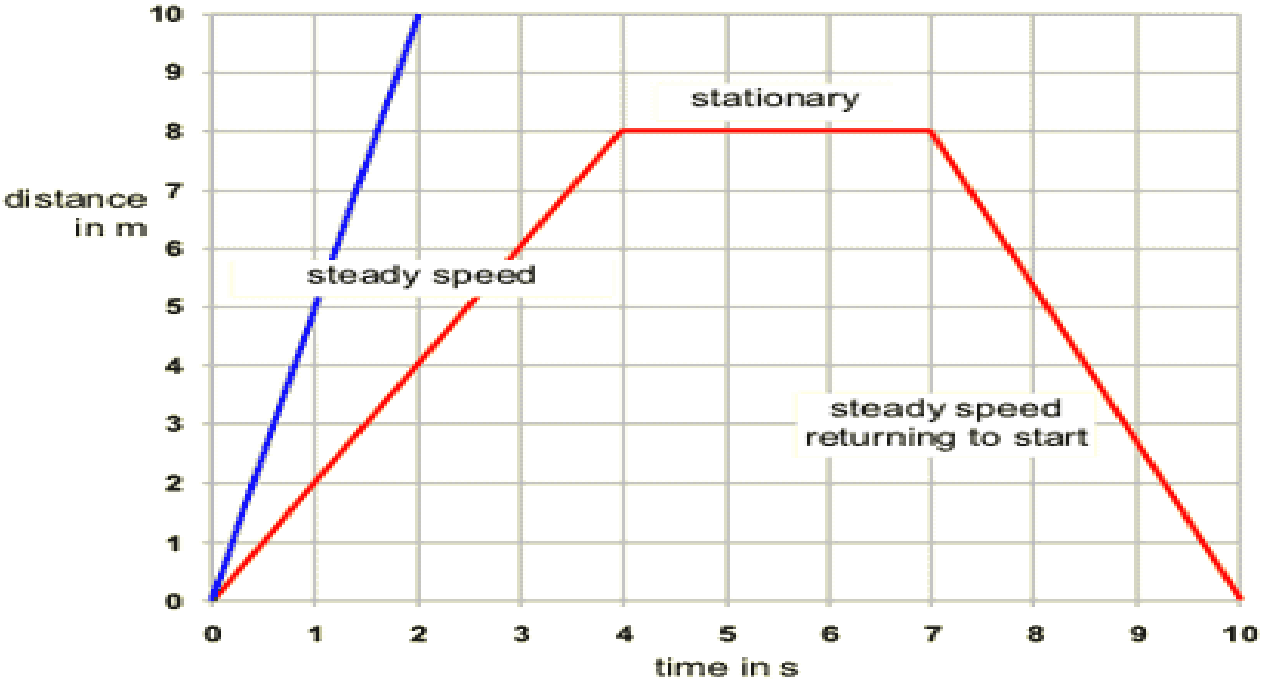
Distance time graph

Graph showing acceleration and deceleration





- Speed from a distance–time graph



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- Acceleration is a vector quantity that is defined as the rate at which an object changes its speed or velocity
- vector quantity has both magnitude and direction eg weight, force, acceleration, displacement, velocity
- scalar quantity has only magnitude but has no directions.eg mass, electric charge, speed, distance etc

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- If the object's speed changes by an equal amount in an equal interval of time, it is called a constant or uniform acceleration, while there is non-uniform acceleration if the object's velocity changes by different amounts in equal intervals of time.

Note: the SI unit of Acceleration is in M/S^2

Every object that falls towards the ground has a constant

acceleration of 9.8 M/S^2

fact

- A rocket or other craft must reach a very high speed or velocity in order to stop falling back to the earth.
- The minimum speed which an object must attain in order to prevent its fall back to the earth is known as the escape velocity.
- For instance for the earth, the escape velocity is about 7 miles/sec

calculations

1. An air craft accelerates from 100m/s to 300m/s in 100sec. What was its acceleration?
2. What is the average acceleration of a car with a velocity which changes from 10km/h to 40 km/h in 6 seconds?